

# **CSM305 – Lab 1: Network Reconnaissance with Nmap and Zen map**

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## **1. Objective**

The objective of this lab was to perform network reconnaissance within a controlled virtual environment. The goal was to identify active hosts, detect open ports, determine running services, and analyze potential security risks using Nmap and Zen map.

## **2. Lab Environment**

The assessment was conducted in a controlled virtual lab environment using VirtualBox. All systems operated within an internal subnet in the **192.168.x.x/24 range**.

### **System Identification (Sanitized):**

- **Target System A** – Windows Server
- **Target System B** – Windows Workstation
- **Attacker Machine** – Kali Linux

All activities were performed within an isolated educational environment. No external systems were accessed.

## **3. Methodology**

The following steps were performed during the assessment:

- Network identification using ip a and ip r
- Connectivity verification using ping
- Host discovery using Nmap
- Port scanning (basic and full scan)

- Service and version detection
- Operating system detection
- Validation using Zenmap

## 4. Connectivity Verification

Connectivity between systems was verified using the ping command. All systems responded successfully, confirming communication within the subnet.

```

kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help

zsh: corrupt history file /home/jasmine/.zsh_history
kali)~]
$ ping -c 4 192.168.1.100
PING 192.168.1.100 (192.168.1.100) 56(84) bytes of data.
64 bytes from 192.168.1.100: icmp_seq=1 ttl=128 time=5.16 ms
64 bytes from 192.168.1.100: icmp_seq=2 ttl=128 time=33.0 ms
64 bytes from 192.168.1.100: icmp_seq=3 ttl=128 time=20.1 ms
64 bytes from 192.168.1.100: icmp_seq=4 ttl=128 time=21.5 ms

--- 192.168.1.100 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3010ms
rtt min/avg/max/mdev = 5.157/19.939/32.972/9.885 ms

kali)~]
$ ping -c 4 192.168.1.100
PING 192.168.1.100 (192.168.1.100) 56(84) bytes of data.
64 bytes from 192.168.1.100: icmp_seq=1 ttl=128 time=8.24 ms
64 bytes from 192.168.1.100: icmp_seq=2 ttl=128 time=2.19 ms
64 bytes from 192.168.1.100: icmp_seq=3 ttl=128 time=2.19 ms
64 bytes from 192.168.1.100: icmp_seq=4 ttl=128 time=2.64 ms

--- 192.168.1.100 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3014ms
rtt min/avg/max/mdev = 2.185/3.811/8.235/2.560 ms

kali)~]
$ ping -c 4 192.168.1.100
PING 192.168.1.100 (192.168.1.100) 56(84) bytes of data.
64 bytes from 192.168.1.100: icmp_seq=1 ttl=64 time=0.025 ms
64 bytes from 192.168.1.100: icmp_seq=2 ttl=64 time=0.049 ms
64 bytes from 192.168.1.100: icmp_seq=3 ttl=64 time=0.132 ms
64 bytes from 192.168.1.100: icmp_seq=4 ttl=64 time=0.057 ms

--- 192.168.1.100 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3071ms
rtt min/avg/max/mdev = 0.025/0.065/0.132/0.040 ms

kali)~]
$ ping -c 4 192.168.1.100
PING 192.168.1.100 (192.168.1.100) 56(84) bytes of data.
64 bytes from 192.168.1.100: icmp_seq=1 ttl=128 time=25.9 ms
64 bytes from 192.168.1.100: icmp_seq=2 ttl=128 time=4.40 ms
64 bytes from 192.168.1.100: icmp_seq=3 ttl=128 time=3.00 ms
64 bytes from 192.168.1.100: icmp_seq=4 ttl=128 time=1.72 ms

--- 192.168.1.100 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3010ms

```

Figure 1: Connectivity Verification using Ping

## 5. Host Discovery

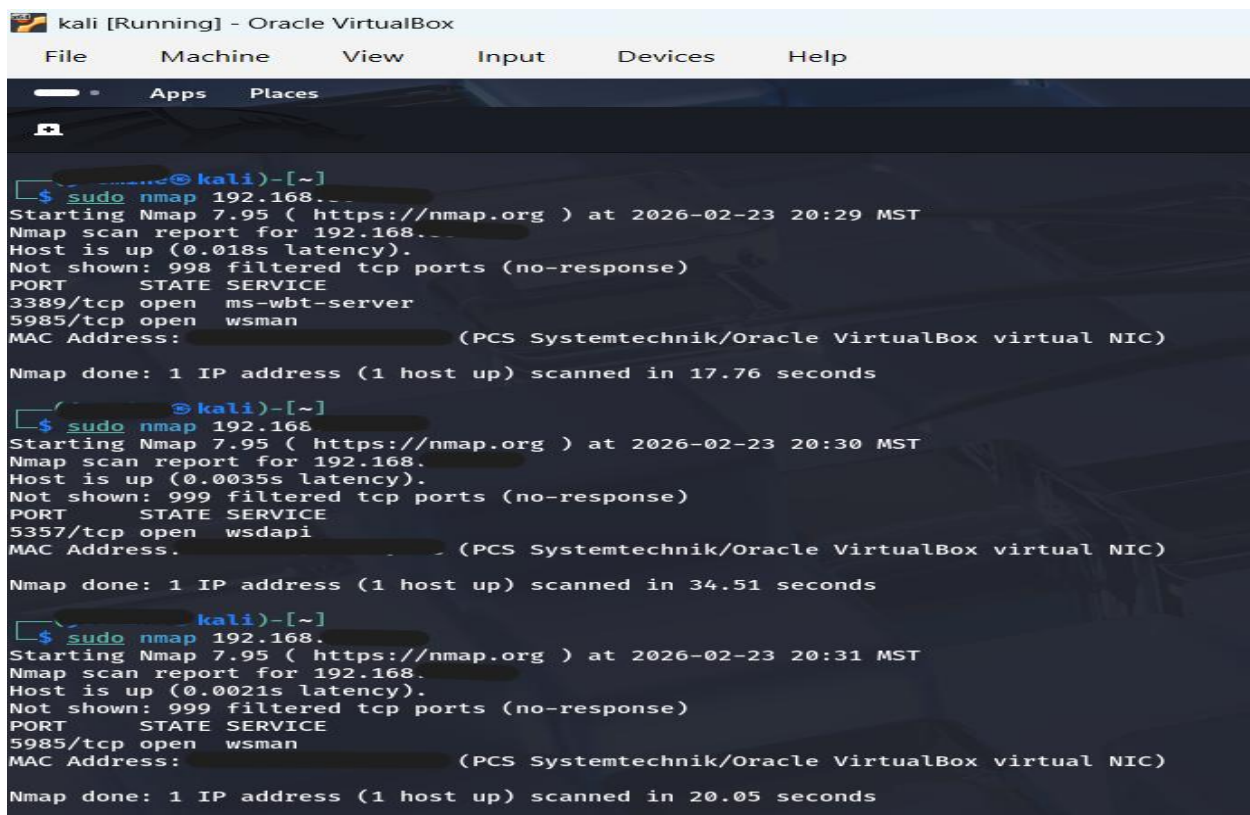
Host discovery was performed using the following command:

```
sudo nmap -sn 192.168.x.x/24
```

**Results:**

| System           | Status | Description               |
|------------------|--------|---------------------------|
| Target System A  | Up     | Windows Server            |
| Target System B  | Up     | Windows Workstation       |
| Attacker Machine | Up     | Kali Linux                |
| Additional Host  | Up     | Virtual Network Interface |

The scan successfully identified multiple active hosts within the internal subnet.



```
kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Apps Places

root@kali:~# sudo nmap 192.168.x.x/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-23 20:29 MST
Nmap scan report for 192.168.x.x
Host is up (0.018s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
3389/tcp  open  ms-wbt-server
5985/tcp  open  wsman
MAC Address: 08:00:27:3C:8E:3E (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 17.76 seconds

root@kali:~# sudo nmap 192.168.x.x/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-23 20:30 MST
Nmap scan report for 192.168.x.x
Host is up (0.0035s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
5357/tcp  open  wsapi
MAC Address: 08:00:27:3C:8E:3E (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 34.51 seconds

root@kali:~# sudo nmap 192.168.x.x/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-23 20:31 MST
Nmap scan report for 192.168.x.x
Host is up (0.0021s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
5985/tcp  open  wsman
MAC Address: 08:00:27:3C:8E:3E (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 20.05 seconds
```

**Figure 2: Host Discovery Scan using Nmap**

## 6. Port Scanning

Basic port scans were conducted on the target systems using:

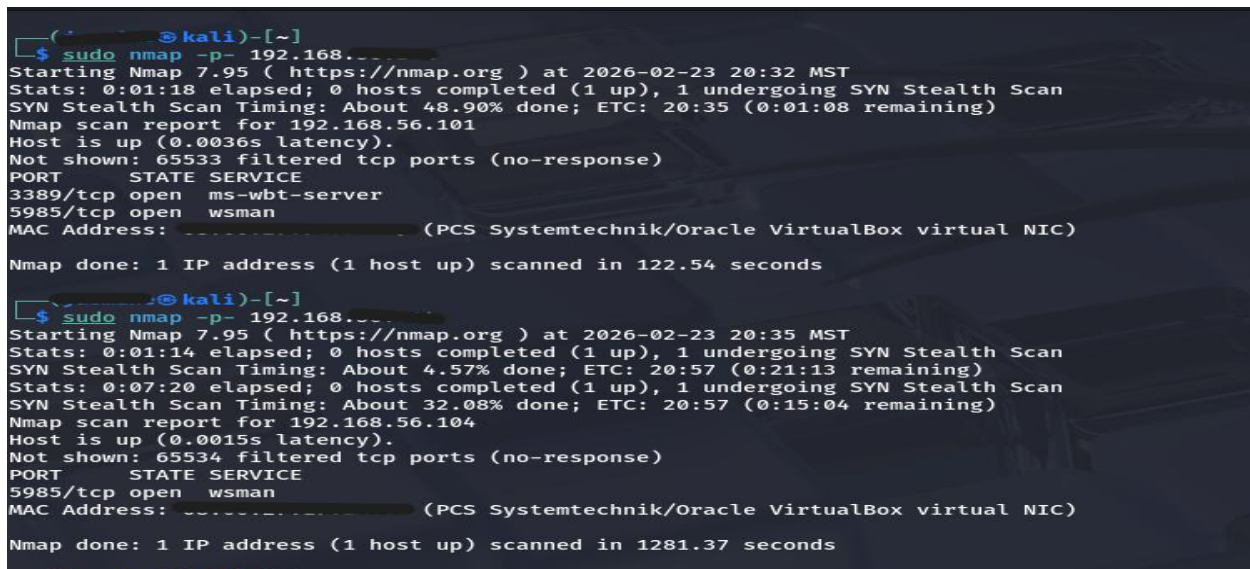
```
sudo nmap 192.168.x.x
```

A full port scan was performed on Target System A using:

```
sudo nmap -p- 192.168.x.x
```

### Findings:

The scans revealed a limited number of open ports and a high number of filtered ports. The presence of filtered ports indicates that firewall rules are actively restricting network access.



```
(kali)-[~]
$ sudo nmap -p- 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-23 20:32 MST
Stats: 0:01:18 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 48.90% done; ETC: 20:35 (0:01:08 remaining)
Nmap scan report for 192.168.56.101
Host is up (0.0036s latency).
Not shown: 65533 filtered tcp ports (no-response)
PORT      STATE SERVICE
3389/tcp  open  ms-wbt-server
5985/tcp  open  wsman
MAC Address: 08:00:27:00:00:00 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 122.54 seconds

(kali)-[~]
$ sudo nmap -p- 192.168.56.104
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-23 20:35 MST
Stats: 0:01:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 4.57% done; ETC: 20:57 (0:21:13 remaining)
Stats: 0:07:20 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 32.08% done; ETC: 20:57 (0:15:04 remaining)
Nmap scan report for 192.168.56.104
Host is up (0.0015s latency).
Not shown: 65534 filtered tcp ports (no-response)
PORT      STATE SERVICE
5985/tcp  open  wsman
MAC Address: 08:00:27:00:00:00 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1281.37 seconds
```

Figure 3: Port Scan Results

## 7. Service and Version Detection

Service detection was performed using:

```
sudo nmap -sV 192.168.x.x
```

### Results:

| System          | Port     | Service           | Description                       |
|-----------------|----------|-------------------|-----------------------------------|
| Target System A | 3389/tcp | ms-wbt-server     | Remote Desktop (RDP)              |
| Target System A | 5985/tcp | Microsoft HTTPAPI | Windows Remote Management (WinRM) |
| Target System B | 5357/tcp | Microsoft HTTPAPI | Device Discovery                  |

These services indicate the presence of remote administrative capabilities and network discovery functions.

```
(jasmine@kali)-[~]
└─$ sudo nmap -sV 192.168.
[sudo] password for jasmine:
Sorry, try again.
[sudo] password for jasmine:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-23 21:03 MST
Nmap scan report for 192.168.
Host is up (0.033s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
5985/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
MAC Address: (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 27.83 seconds

(jasmine@kali)-[~]
└─$ sudo nmap -sV 192.168.
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-23 21:04 MST
Nmap scan report for 192.168.
Host is up (0.0070s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
5357/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
MAC Address: (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 37.74 seconds

(jasmine@kali)-[~]
└─$
```

**Figure 4: Service Detection using Nmap**

## 8. Operating System Detection

Operating system detection was performed using: *sudo nmap -O 192.168.x.x*

The results indicated that both systems are Microsoft Windows-based with high confidence. However, exact OS versions could not be confirmed due to limited scan responses caused by filtered ports.

```
kali:~$ sudo nmap -O 192.168.1.10
Starting Nmap 7.95 ( https://nmap.org ) at 2020-02-23 21:07 MST
Nmap scan report for 192.168.1.10
Host is up (0.013s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
3389/tcp  open  ms-wbt-server
5985/tcp  open  wsman
MAC Address: 08:00:27:00:00:00 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
OS fingerprint not ideal because: Missing a closed TCP port so results incomplete
No OS matches for host
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 21.59 seconds

kali:~$ sudo nmap -O 192.168.1.10
Starting Nmap 7.95 ( https://nmap.org ) at 2020-02-23 21:09 MST
Nmap scan report for 192.168.1.10
Host is up (0.0035s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
5357/tcp  open  wsddapi
MAC Address: 08:00:27:00:00:00 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose phone
Running (JUST GUESSING): Microsoft Windows 11(10)/2022(2008)/Phone (96%)
OS CPE: cpe:/o:microsoft:windows_11 cpe:/o:microsoft:windows_10 cpe:/o:microsoft:windows_server_2022 cpe:/o:microsoft:windows_server_2008:sp1 cpe:/o:microsoft:windows
Aggressive OS guesses: Microsoft Windows 11 21H2 (96%), Microsoft Windows 10 (91%), Microsoft Windows 10 1607 (91%), Microsoft Windows Server 2022 (90%), Microsoft Windows Server 2008 SP1 (88%), Microsoft Windows Phone 7.5 or 8.0 (86%), Microsoft Windows 10 1511 - 1607 (86%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 30.33 seconds
```

Figure 5: Operating System Detection

## 9. Zenmap Analysis

Zenmap was used to perform graphical scans, including Ping Scan and Intense Scan profiles. The results confirmed findings from command-line scans, including active hosts and exposed services.

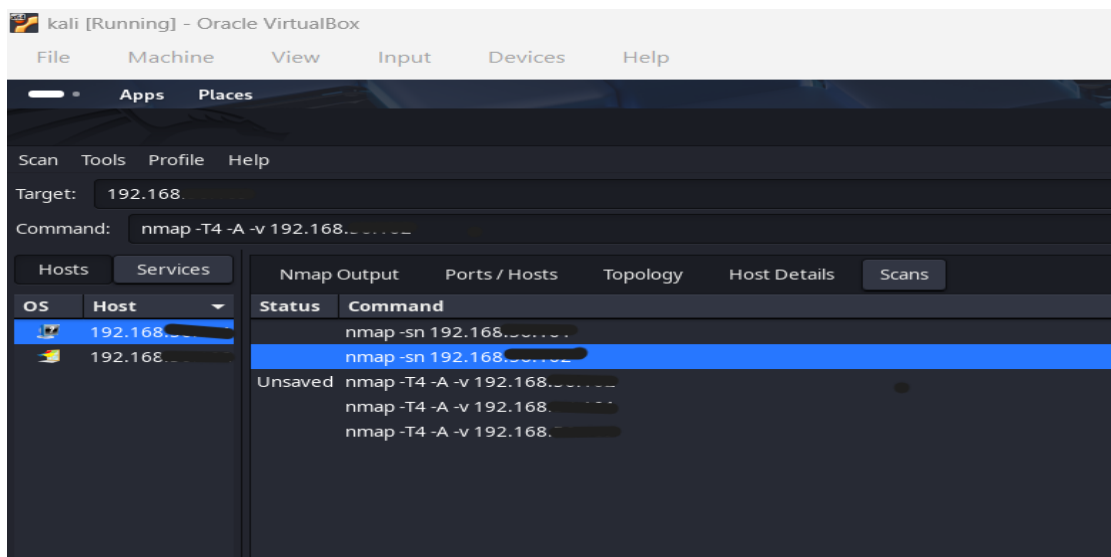


Figure 6: Zenmap Scans

## 10. Port State Analysis

- **Open** – A service is actively listening and responding
- **Closed** – No service is running, but the system responds
- **Filtered** – No response is received, typically due to firewall filtering

The large number of filtered ports suggests strong host-based firewall protections.

## 11. Technical Findings Summary

- Multiple active systems were identified within the subnet
- Target System A exposes:
  - Remote Desktop (3389)
  - Windows Remote Management (5985)
- Target System B exposes:
  - Port 5357 (device discovery)
- The majority of ports on both systems are filtered
- Systems were identified as Windows-based, but exact versions were not confirmed

## 12. Risk Analysis

### Target System A (Windows Server)

- **Impact:** High
- **Likelihood:** Medium

Remote administrative services such as RDP and WinRM increase the attack surface and are common targets for credential-based attacks. If compromised, these services could allow unauthorized access and control.

### Target System B (Windows Workstation)

- **Impact:** Medium

- **Likelihood:** Low to Medium

The device discovery service increases internal network visibility and may support lateral movement if exploited.

## **13. Recommendations**

### **Immediate Actions**

- Restrict access to ports 3389 and 5985 using firewall rules
- Disable unnecessary remote services

### **Security Enhancements**

- Enforce strong password policies
- Enable account lockout mechanisms
- Implement Multi-Factor Authentication (MFA)

### **Long-Term Improvements**

- Configure WinRM to use HTTPS instead of HTTP
- Apply strong encryption for RDP connections
- Implement network segmentation
- Enable centralized logging and monitoring

## **14. Ethical Considerations**

This assessment was conducted within a controlled educational lab environment. All IP addresses, system identifiers, and file paths have been anonymized or generalized to maintain security and privacy best practices.



## 15. Conclusion

This lab demonstrated how network reconnaissance techniques can be used to identify active hosts, open ports, and exposed services within a controlled environment. While firewall protections reduced overall exposure, critical administrative services remained accessible, highlighting potential security risks. The assessment emphasizes the importance of securing remote access services and implementing strong authentication and monitoring controls.

## 16. Appendix

### Commands Used

```
ip a
```

```
ip r
```

```
sudo nmap -sn 192.168.x.x/24
```

```
sudo nmap 192.168.x.x
```

```
sudo nmap -p- 192.168.x.x
```

```
sudo nmap -sV 192.168.x.x
```

```
sudo nmap -O 192.168.x.x
```